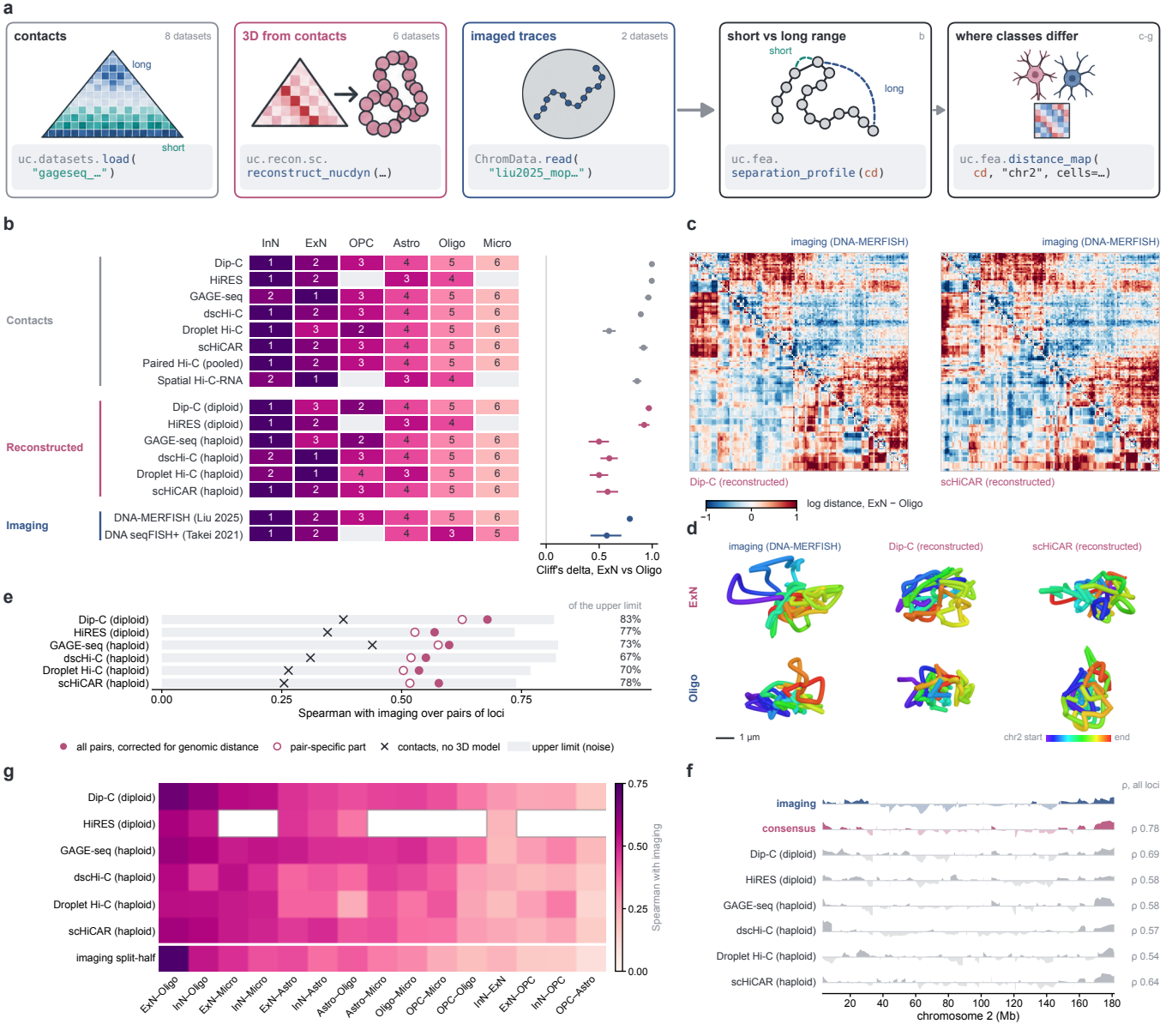

a
contacts
8 datasets
long
short
uc.datasets.load("gageseq_…")
3D from contacts
6 datasets
uc.recon.sc.reconstruct_nucdyn(…)
imaged traces
2 datasets
ChromData.read("liu2025_mop…")
short vs long range
b
short
long
uc.fea.separation_profile(cd)
where classes differ
c-g
uc.fea.distance_map(cd, "chr2", cells=…)
b
InN ExN OPC Astro Oligo Micro
Contacts
Dip-C 1 2 3 4 5 6
HiRES 1 2 3 4
GAGE-seq 2 1 3 4 5 6
dscHi-C 1 2 3 4 5 6
Droplet Hi-C 1 3 2 4 5 6
scHiCAR 1 2 3 4 5 6
Paired Hi-C (pooled) 1 2 3 4 5 6
Spatial Hi-C-RNA 2 1 3 4
Reconstructed
Dip-C (diploid) 1 3 2 4 5 6
HiRES (diploid) 1 2 3 4
GAGE-seq (haploid) 1 3 2 4 5 6
dscHi-C (haploid) 2 1 3 4 5 6
Droplet Hi-C (haploid) 2 1 4 3 5 6
scHiCAR (haploid) 1 2 3 4 5 6
Imaging
DNA-MERFISH (Liu 2025) 1 2 3 4 5 6
DNA seqFISH+ (Takei 2021) 1 2 4 3 5
0.0 0.5 1.0
Cliff's delta, ExN vs Oligo
c
imaging (DNA-MERFISH)
imaging (DNA-MERFISH)
Dip-C (reconstructed)
scHiCAR (reconstructed)
log distance, ExN − Oligo
−1 0 1
d
imaging (DNA-MERFISH)
Dip-C (reconstructed)
scHiCAR (reconstructed)
ExN
Oligo
1 µm
chr2 start
end
e
of the upper limit
Dip-C (diploid) 83%
HiRES (diploid) 77%
GAGE-seq (haploid) 73%
dscHi-C (haploid) 67%
Droplet Hi-C (haploid) 70%
scHiCAR (haploid) 78%
0.00 0.25 0.50 0.75
Spearman with imaging over pairs of loci
all pairs, corrected for genomic distance
pair-specific part
contacts, no 3D model
upper limit (noise)
g
Dip-C (diploid)
HiRES (diploid)
GAGE-seq (haploid)
dscHi-C (haploid)
Droplet Hi-C (haploid)
scHiCAR (haploid)
imaging split-half
0.75 0.50 0.25 0.00
Spearman with imaging
ExN–Oligo InN–Oligo ExN–Micro InN–Micro ExN–Astro InN–Astro Astro–Oligo Astro–Micro Oligo–Micro OPC–Micro OPC–Oligo InN–ExN ExN–OPC InN–OPC OPC–Astro
f
ρ, all loci
imaging
consensus ρ 0.78
Dip-C (diploid) ρ 0.69
HiRES (diploid) ρ 0.58
GAGE-seq (haploid) ρ 0.58
dscHi-C (haploid) ρ 0.57
Droplet Hi-C (haploid) ρ 0.54
scHiCAR (haploid) ρ 0.64
20 40 60 80 100 120 140 160 180
chromosome 2 (Mb)